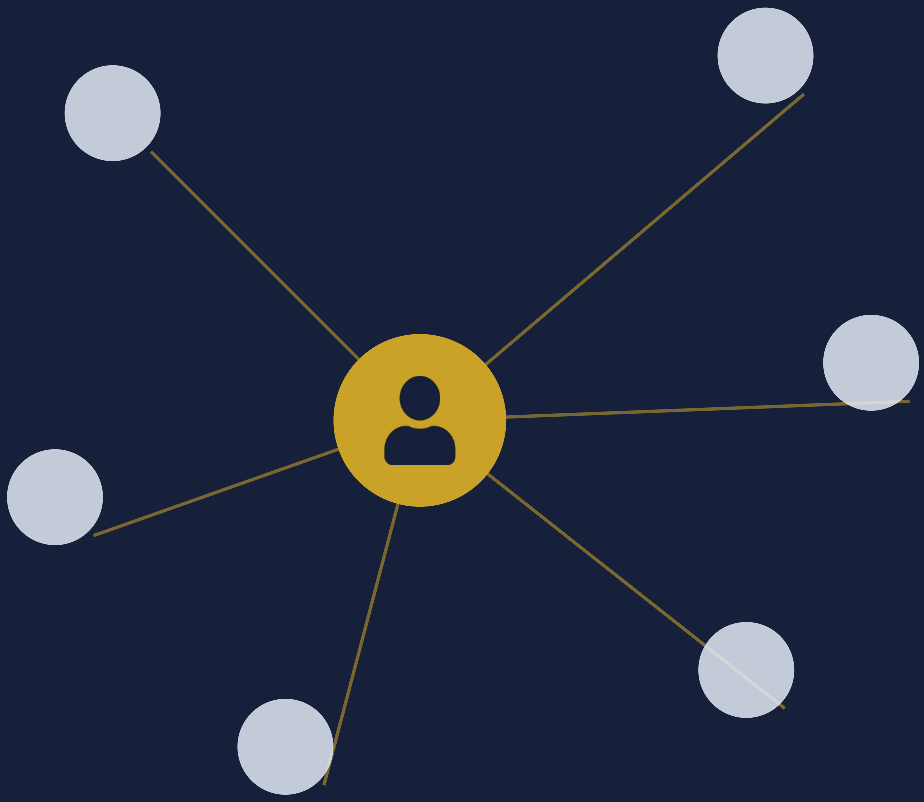
E-COMMERCE · THE BUSINESS CASE

The Business Case for Product Recommendations

Turning a vast catalog into a personal storefront

Projected impact of integrating recommendations into our platform



Presented by Your Name · June 2026

EXPECTED IMPACT AT A GLANCE

What integrating recommendations should do for our platform

17%

of global online orders are driven by AI recommendations

Salesforce, 2024

5–15%

typical revenue lift from personalization

McKinsey, 2021

+7%

average order value with AI-driven personalization

Salesforce, 2024

Modeled on $100M annual revenue (illustrative)

$5M – $15M added revenue per year

Swap $100M for our actual revenue — the % ranges are sourced.

Where it shows up

Conversion ↑ | Order value ↑ | Retention ↑

Proven at scale: Amazon, Taobao / Alibaba, and JD.com run recommendations as core infrastructure — detailed later in this deck.

Sources: Salesforce Shopping Index (2024); McKinsey & Company (2021). $100M base is illustrative — replace with your revenue.

WHY WE NEED THIS

A bigger catalog than anyone can browse — and rising expectations

Discovery is the bottleneck

Most of our catalog is never seen; shoppers can't buy what they never find.

Choice overload stalls purchases

Endless undifferentiated options delay decisions and depress conversion.

Personalization is now expected

71% of consumers expect personalized interactions; 76% are frustrated when it's missing.

What AI recommendations drive today

17%

of global online orders driven by AI — recommendations, promotions & service

+7%

higher average order value for retailers using generative AI

Salesforce Shopping Index, 2024 (1.5B shoppers).
Recommendations are how leaders convert a huge catalog into sales.

Sources: Salesforce Shopping Index (2024); McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying" (2021).

HOW MUCH COULD IT MOVE REVENUE?

Published lift ranges, applied transparently to our numbers

What the research shows

- Personalization most often lifts revenue 5–15% (sector range up to ~25%)
- Can reduce customer-acquisition cost by up to 50%
- Improves marketing ROI by 10–30%
- Fast-growing firms earn ~40% more of their revenue from personalization

All figures: McKinsey (2021).

Illustrative on $100M annual revenue

Applying the 5–15% revenue-lift range:

5% lift	10% lift	15% lift
$5M /year	$10M /year	$15M /year

Replace $100M with our actual revenue. The percentages are sourced; the base is a placeholder for modeling, not a forecast.

Source: McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021.

FROM RELEVANCE TO REVENUE

How recommendations reach the numbers we report

Relevant recommendations → Deeper engagement & discovery →

Conversion rate up

More sessions end in a purchase

Average order value up

Cross-sell raises basket size

Retention up

A tailored store earns repeat visits

Net effect: higher revenue per visitor and stronger lifetime value.

WHERE CUSTOMERS SEE IT

Five surfaces, each with a business job

Personalized feed

Drives engagement

“For you” home and category feeds

Frequently bought together

Grows basket size

Cross-sell at cart & product page

Trending & popular

Converts undecided shoppers

Strong default with little data

Similar items

Extends the session

“More like this” keeps browsing

Recently viewed

Recovers lost intent

Easy path back to consideration

HOW IT WORKS (IN PLAIN TERMS)

Learn from behavior, predict relevance, rank

The system learns from what shoppers browse and buy, predicts what each person is most likely to want next, and shows those items first — automatically, for every visitor.

What we know

Shopper history, product details, and the moment (device, time).

Smart matching

Scores how relevant each product is to this shopper.

Best picks shown

A short, ranked list on the page — refreshed continuously.

PROOF AT SCALE

Market leaders treat recommendations as core infrastructure

Amazon

Pioneered item-to-item recommendations ("customers who bought this also bought"), launched 1998 and published in IEEE Internet Computing.

A foundational cross-sell engine, copied across the web.

Taobao / Alibaba

"Guess What You Like" home feed uses graph embeddings (EGES) to personalize a billion-item catalog; A/B-tested CTR gains.

Behavior-driven discovery at marketplace scale.

JD.com (Jingdong)

Deep-learning semantic retrieval and re-ranking serve hundreds of millions of users at ~$100B+ annual GMV.

Relevance tied directly to sales at huge scale.

Sources: Smith & Linden, IEEE Internet Computing (2017); Wang et al., KDD 2018; Li et al. (JD.com), 2021.

SWOT ANALYSIS

Adopting recommendations on our platform

Strengths

- We already own first-party behavioral data — the fuel for recommendations
- Scales personalization to every visitor automatically
- Mature, proven technology with vendor and open-source options
- Directly targets conversion, AOV, and retention

Weaknesses

- Cold start: thin signal for new users and items at launch
- Needs data quality, ML talent, and low-latency serving
- Results depend on traffic volume — thin data, weaker lift
- Ongoing maintenance and experimentation overhead

Opportunities

- 5–15% revenue lift typical (McKinsey, 2021)
- Differentiation: 71% of consumers expect personalization
- Cross-sell and surfacing of long-tail catalog
- Lower acquisition cost (up to −50%) via better retention

Threats

- Privacy regulation and consent (e.g., GDPR, CCPA)
- Done poorly, personalization can cut revenue ~5% (McKinsey)
- Competitors with mature systems raise the bar
- Bias, filter bubbles, and reputational risk if unmanaged

Source: McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021.

KEY RISKS & HOW WE MANAGE THEM

What we'll actively control as we scale

Filter bubbles

Over-narrowing hurts discovery.

Mitigation: Inject diversity / variety controls.

Popularity bias

Best-sellers crowd out the long tail.

Mitigation: Debias training; promote fresh items.

Privacy & consent

Regulatory and trust exposure.

Mitigation: Consent-aware, data-minimized design.

Fairness

Uneven exposure across groups/sellers.

Mitigation: Audit exposure parity regularly.

Poor personalization

Done wrong, it can cut revenue ~5%.

Mitigation: Guardrails + A/B test every change.

Brand safety

Off-brand or non-compliant results.

Mitigation: Business rules and human review.

Source: McKinsey & Company, "The value of getting personalization right—or wrong—is multiplying," 2021.

A PHASED INVESTMENT

Prove value early, then scale what works

Phase 1 — Crawl

0–3 months

Popularity, trending, and rules-based “bought together.”

Quick wins, low cost

Phase 2 — Walk

3–9 months

Add behavioral collaborative filtering and similar-item recall.

Personalization begins

Phase 3 — Run

9–18 months

Personalized feeds, hybrid models, real-time ranking.

Compounding returns

Each phase ships value on its own and gates the next on measured lift. The next slide shows a real platform that followed this exact path.

REAL-WORLD PRECEDENT

How Amazon actually walked this path

Amazon's own published history maps directly onto the crawl → walk → run plan.

1 Pre-1998

2 1998

3 2003

4 2017+

Crawl

Editorial picks & popularity

Early Amazon surfaced best-sellers and hand-curated lists across a mostly-books catalog.

Walk

Item-to-item collaborative filtering

Launched “customers who bought this also bought,” computing item-item similarities offline for fast lookups.

Walk

Published & proven to scale

Detailed in IEEE Internet Computing; scaled where user-based methods couldn't, and spread across the web.

Run

A store for every customer

Two decades of refinement: the whole store personalized per shopper, continuously A/B tested.

Sources: Smith & Linden, “Two Decades of Recommender Systems at Amazon.com,” IEEE Internet Computing (2017); Linden, Smith & York, IEEE Internet Computing (2003).

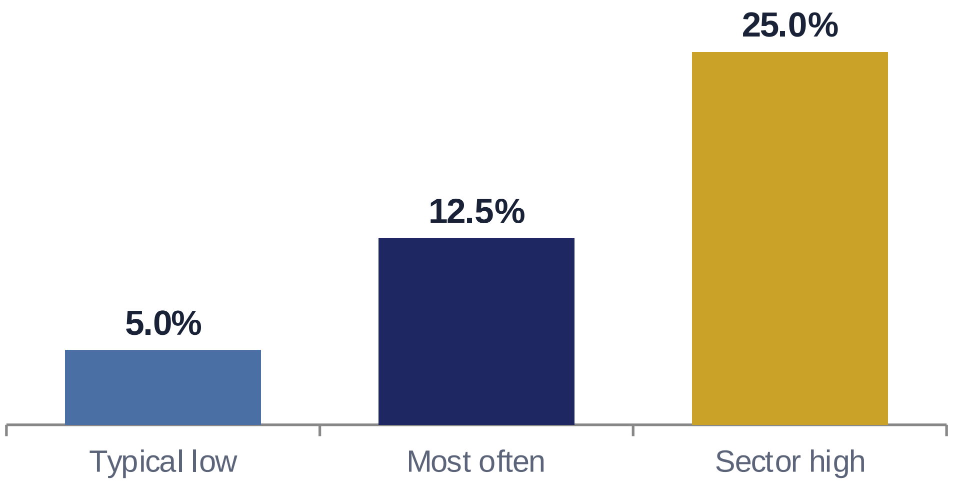
HOW WE'LL PROVE ROI

No claim ships without a controlled test

The method

1. Show recommendations to a random half of traffic (A/B test)
2. Compare against the holdout over a fixed window
3. Measure conversion, AOV, and revenue per session
4. Roll out only if lift is positive and statistically sound

What “good” looks like (published benchmark)



Personalization revenue-lift range — McKinsey (2021): most often 10–15%, sector range 5–25%.

Source: McKinsey & Company, “The value of getting personalization right—or wrong—is multiplying,” 2021.

THE ASK

What we'd like to move forward

Approve a Phase 1 pilot

A scoped, time-boxed test on one or two high-traffic surfaces.

Fund the team & tooling

A small cross-functional team plus experimentation tooling.

Agree success criteria

A clear lift target (benchmark: McKinsey 5–15%) that gates Phase 2.

KEY TAKEAWAYS

What to carry out of this session

1. Recommendations turn a vast catalog into a personal storefront
2. Published research: personalization typically lifts revenue 5–15% (McKinsey)
3. Market leaders (Amazon, Taobao, JD.com) treat them as core infrastructure
4. Phase the investment along Amazon's proven crawl → walk → run path
5. Prove every claim with A/B tests; manage privacy, bias, and fairness

Let's approve a focused pilot and measure the lift.

Thank you

SOURCES & REFERENCES

Every figure in this deck is publicly sourced

Salesforce Shopping Index (2024)	Holiday-peak analysis of 1.5B shoppers. — AI (incl. personalized recommendations) drove 17% of global online orders; +7% AOV for retailers using generative AI. *salesforce.com/news/stories/ai-for-holiday-shopping-predictions*
McKinsey & Company (2021)	“The value of getting personalization right—or wrong—is multiplying.” — 5–15% revenue lift; CAC up to −50%; marketing ROI +10–30%; 71%/76% expectation; ~5% downside if done poorly. *mckinsey.com/capabilities/growth-marketing-and-sales/our-insights*
Gomez-Uribe & Hunt (2015)	“The Netflix Recommender System: Algorithms, Business Value, and Innovation,” ACM TMIS. — Recommendations estimated worth ~$1B/year via reduced churn. *ACM Transactions on Management Information Systems, 6(4)*
Smith & Linden (2017); Linden et al. (2003)	“Two Decades of Recommender Systems at Amazon.com” and “Amazon.com Recommendations: Item-to-Item Collaborative Filtering,” IEEE Internet Computing. *ieeexplore.ieee.org/document/7927889*
Wang et al. (2018)	“Billion-scale Commodity Embedding for E-commerce Recommendation in Alibaba,” KDD 2018 — Taobao EGES graph embeddings; A/B-tested CTR gains. *arxiv.org/abs/1803.02349*
JD.com — Li et al. (2021); Zhao et al. (2017)	“From Semantic Retrieval to Pairwise Ranking”; JD scale (hundreds of millions of users, ~$100B GMV). *arxiv.org/abs/2103.12982 · arxiv.org/abs/1709.00300*

APPENDIX

Platform deep-dive

Development history (pre- and post-GPT) and effectiveness by user class — Amazon, Taobao / Alibaba, and JD.com. All figures cited from sources published in 2024–2026.

Amazon — recommendation system history

From item-to-item collaborative filtering to a generative AI shopping assistant.

PRE-GPT ERA

1998
Item-to-item CF
"Customers who bought this also bought."

2003
Published (IEEE)
Offline similarity → fast lookups; scales.

2017
Whole-store personalization
Deep learning across the catalog.

GPT / LLM ERA

2024
Rufus + COSMO
GenAI assistant on Bedrock (RAG) + commonsense knowledge graph.

2025
Rufus at scale
Cross-ecosystem memory; per-shopper answers.

GPT-era impact

~$12B
incremental annualized sales (Amazon Q4 2025)

300M+
Rufus users · 149% YoY active-user growth

+60%
purchase completion among Rufus-engaged shoppers

Sources: Amazon Science (2024); Amazon Q4 2025 earnings; COSMO — ACM SIGMOD 2024; Linden et al., IEEE Internet Computing (2003, historical).

APPENDIX · PLATFORM DEEP-DIVE

Taobao / Alibaba — recommendation system history

From graph embeddings to LLM-based, intent-centric generative recommendation.

PRE-GPT ERA

GPT / LLM ERA

2018	2018	2025	2025	2025
EGES graph embeddings	**Deep Interest Network**	**LUM — Large User Model**	**URM universal retriever**	**RecGPT**
Powers “Guess What You Like.” (KDD 2018)	Attention-based CTR modeling (KDD 2018).	Scaling-law user model; Taobao search A/B gains.	LLM retrieval; tens-of-ms latency.	Intent-centric LLM recommendation framework.

GPT-era impact

+2.9% / +1.2%

CTR / RPM in Taobao search A/B — LUM (2025)

18.8% → 46.2%

new-category discovery via serendipity retrieval — URM (2025)

Sources: Wang et al., KDD 2018 (EGES); Zhou et al., KDD 2018 (DIN); Yan et al., 2025 (LUM); Jiang et al., 2025 (URM, arXiv:2502.03041); Yi et al., 2025 (RecGPT).

JD.com (Jingdong) — recommendation system history

From deep-learning retrieval to generative CTR and generative retrieval in production.

PRE-GPT ERA

GPT / LLM ERA

2017	2021	2025	2025	2025
Telepath	**Semantic retrieval + re-rank**	**GenCTR**	**GRAM**	**CFRAG**
Deep-learning, vision-aware recommender.	Deep retrieval with pairwise ranking for search.	Generative CTR for search ads (gen→discriminative).	Generative retrieval & alignment for e-commerce.	LLM-embedding-augmented recall (RAG for rec).

GPT-era impact

Generative CTR

generative→discriminative transfer in production (2025)

Generative retrieval

GRAM & CFRAG bring LLM retrieval to JD search (2025)

Sources: Zhao et al. (Telepath), 2017; Li et al., 2021; Kong et al., 2025 (GenCTR, arXiv:2507.11246); JD GRAM & CFRAG, 2025.

Effectiveness by user class (LLM era)

Recent studies show the generative/LLM shift helps most where classical methods struggle — new and cold-start users.

New & cold-start

Little or no history

- **Alibaba URM:** new-category discovery 18.8% → 46.2% (2025)
- **Video platform:** +120% Recall@200 for coldest items; +36% cold-item share (2025)
- **Netflix:** +8% recall on novel-content discovery (2025)

Casual & returning

Sporadic, intent-led visits

- **Amazon Rufus:** per-shopper answers; cross-ecosystem memory (Nov 2025)
- **RecGPT (Taobao):** intent-centric reasoning replaces log-fitting (2025)
- **Conversational search:** natural-language intent matching over keywords

Core & power users

Long, rich histories

- **Amazon Rufus:** +60% purchase completion among engaged shoppers (2025)
- **Alibaba LUM:** +2.9% CTR from scaling the user model (2025)
- **Long-sequence models:** large user models exploit deep history

Sources (2025): Jiang et al. (URM); Yan et al. (LUM); Amazon Q4 2025; Li et al., Nov 2025 (Netflix); cold-start video study, Aug 2025. Platform-level per-segment figures are rarely disclosed; cross-industry studies used as directional evidence.

APPENDIX · SOURCES

LLM-era sources (2024–2026)

Amazon — Rufus	Amazon Science, "The technology behind Amazon's GenAI shopping assistant, Rufus" (2024); scale & sales via Amazon Q4 2025 earnings.	*amazon.science/blog/...-rufus*
Amazon — COSMO	Commonsense knowledge graph built with LLMs, ACM SIGMOD 2024.	*amazon.science (COSMO, SIGMOD 2024)*
Alibaba — URM	Jiang et al., "LLM as Universal Retriever in Industrial-Scale Recommender System" (2025).	*arxiv.org/abs/2502.03041*
Alibaba — LUM & RecGPT	Yan et al., "Large User Model" (2025); Yi et al., "RecGPT Technical Report" (2025).	*arXiv 2025 (LUM); RecGPT report, Aug 2025*
JD.com — GenCTR & GRAM	Kong et al., "Generative CTR Prediction..." (2025); JD "GRAM: Generative Retrieval and Alignment" (2025).	*arxiv.org/abs/2507.11246*
User-class / cold-start	Netflix cold-start (Li et al., Nov 2025); video-platform cold-start study (Aug 2025); cold-start LLM survey (2025).	*arXiv 2025*

Note: pre-GPT milestones (item-to-item CF 1998/2003; EGES & DIN 2018; JD Telepath 2017) are historical; all quantitative effectiveness figures above are from 2024–2026 publications.